MILITARY SPECIFICATION SHEET

INTERCOMMUNICATIONS SETS, AN/AIC-14 and AN/AIC-14A

MICROPHONE MIXING MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the Microphone Mixing Module used in Audio Frequency Amplifier AM-3889/AIC-14A.

1.2 Military Part Number. The military part number shall be M22352/4-1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The Microphone Mixing Module shall provide amplification of two independent audio signals as specified herein when used in audio frequency amplifier AM-3889/AIC-14A. The amplifier(s) shall combine the signals to form a single output.

3.1.1 Item Diagram. The Microphone Mixing Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

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3.1.2 Interface Definition. The Microphone Mixing Module input and output signals will normally display the characteristics presented in Table I:

TABLE I. Signal characteristics

Signal Name	Characteristics
Inputs:	
Microphone No. 1 (MIC NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Microphone No. 2 (MIC NO. 2)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	+27.5 ±0.5Vdc
Outputs:	
Audio (AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Microphone Current (MIC CUR)	50 ±10 milliamperes dc (into a 100 ohm load)
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 Microphone No. 1 (MIC NO. 1). The Microphone Mixing Module shall have a NO. 1 input circuit having the following performance characteristics:

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3.2.1.1.1.1 Signal Frequency. The MIC NO. 1 input shall accept input drive signals which range from 300 to 7,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The MIC NO. 1 input shall accept input drive signals which range in amplitude from 0.01 to 0.10 Vac. The Microphone Mixing Module shall be capable of withstanding overvoltage inputs of up to 1.0 Vac without sustaining damage. The Microphone Mixing Module shall be capable of operating within specification when the MIC NO. 1 signal input contains a dc component of up to 2 Volts.

3.2.1.1.1.3 Input Impedance. The MIC NO. 1 input impedance shall be 100 ohms ±20 percent.

3.2.1.1.2 Microphone No. 2 (MIC NO. 2). The MIC NO. 2 input characteristics shall be as described in Section 3.2.1.1.1 for the MIC NO. 1 input.

3.2.1.1.3 Power (PWR). The Microphone Mixing Module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the Microphone Mixing Module shall not exceed 100 milliamperes.

The Microphone Mixing Module shall contain an insertion filter for the MIC CUR output which shall be capable of attenuating ripple on the +27.5 Vdc line a minimum of 20 dB within the range of 300 and 7000 Hz.

3.2.1.2 Output Characteristics. The Microphone Mixing Module shall have the following output characteristics:

3.2.1.2.1 Microphone Audio (MIC AUDIO). The MIC AUDIO output shall have the following characteristics:

3.2.1.2.1.1 Frequency and Amplitude. The AUDIO signal output amplitude shall equal the input amplitude at 1,000 Hz within ±2dB. Within the frequency range from 300 to 7000 Hz the signal amplitude shall not vary more than 2dB above or below the value at 1,000 Hz when terminated in a 100 ohm resistive load. The input/output characteristic (gain/loss) of each amplifier shall be unity (0 dB) and not vary more than ±2dB when the input is varied from 0.01 to 0.10 volts at 1000 Hz.

3.2.1.2.1.2 Output Impedance. Each amplifier shall be capable of providing rated output when terminated in a 100 ohm noninductive load.

3.2.1.2.1.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7,000 Hz over the amplitude range of 0.01 to 0.10 Vac shall not exceed 5 percent under standard conditions as defined in Section 3.1a of MIL-STD-810. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 8 percent throughout the environmental envelope of MIL-E-5400 Class 2.

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4.5.2 Tests

3.2.1.2.1.4 Output Power. Each amplifier shall be capable of delivering a minimum of 14 milliwatts into a 100 ohm noninductive load.

3.2.1.2.1.5 Amplifier Stability. The amplifier which produces the output signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.1.6 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated under standard conditions as defined in Section 3.1a of MIL-STD-810 with the input circuit shorted to system ground.

3.2.1.2.1.7 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.1.8 Pull-up. The AUDIO output shall function as specified herein when connected to +27.5 ±0.5 Vdc through a low pass filter consisting of 390 ohms in series with 100 ohms with the junction of these resistors shunted to ground through an 80 microfarad capacitor as shown in Figure 3.

3.2.1.2.2 Microphone Current (MIC CUR). The MIC CUR output shall provide 50 ± 10 mA of current to a 100 ohm load. The +27.5 Vdc power input shall be filtered such that incoming ripple is suppressed by at least 20 dB within the range of 300 and 7000 Hz before being output as the MIC CUR output.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The Microphone Mixing Module shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The Microphone Mixing Module shall weigh no more than 3.0 ounces.

3.3 Design and Construction. The design and construction of the Microphone Mixing Module shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

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4.2 Test Equipment. See 4.3.3.1 MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of product. See 4.3.3.2.1 MIL-I-22352(AS).

4.3.2 Operational test. See 4.3.3.2.2 MIL-I-22352(AS).

4.3.2.1 Input Impedance. Verify that the NO. 1 and NO. 2 are in accordance with 3.2.1.1.1.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.1.1 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.1.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and o.10 Vac, verify that the amplifier meets the requirements of 3.2.1.2.1.1.

4.3.2.5 Output Power. With the amplifier HDST HI terminated with 600 ohms, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.2.1.2.1.4 herein.

4.3.2.6 Distortion. Verify the requirements of 3.2.1.2.1.3 herein.

4.3.2.7 Amplifier stability. Verify the requirements of 3.2.1.2.1.5 herein.

4.3.2.8 Internal noise. Verify the requirements of 3.2.1.2.1 herein.

4.3.2.9 Susceptibility to stray magnetic fields. Verify the performance of 3.2.1.2.1.7 herein.

Preparing Activity:
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Project No. 5831-N042-4

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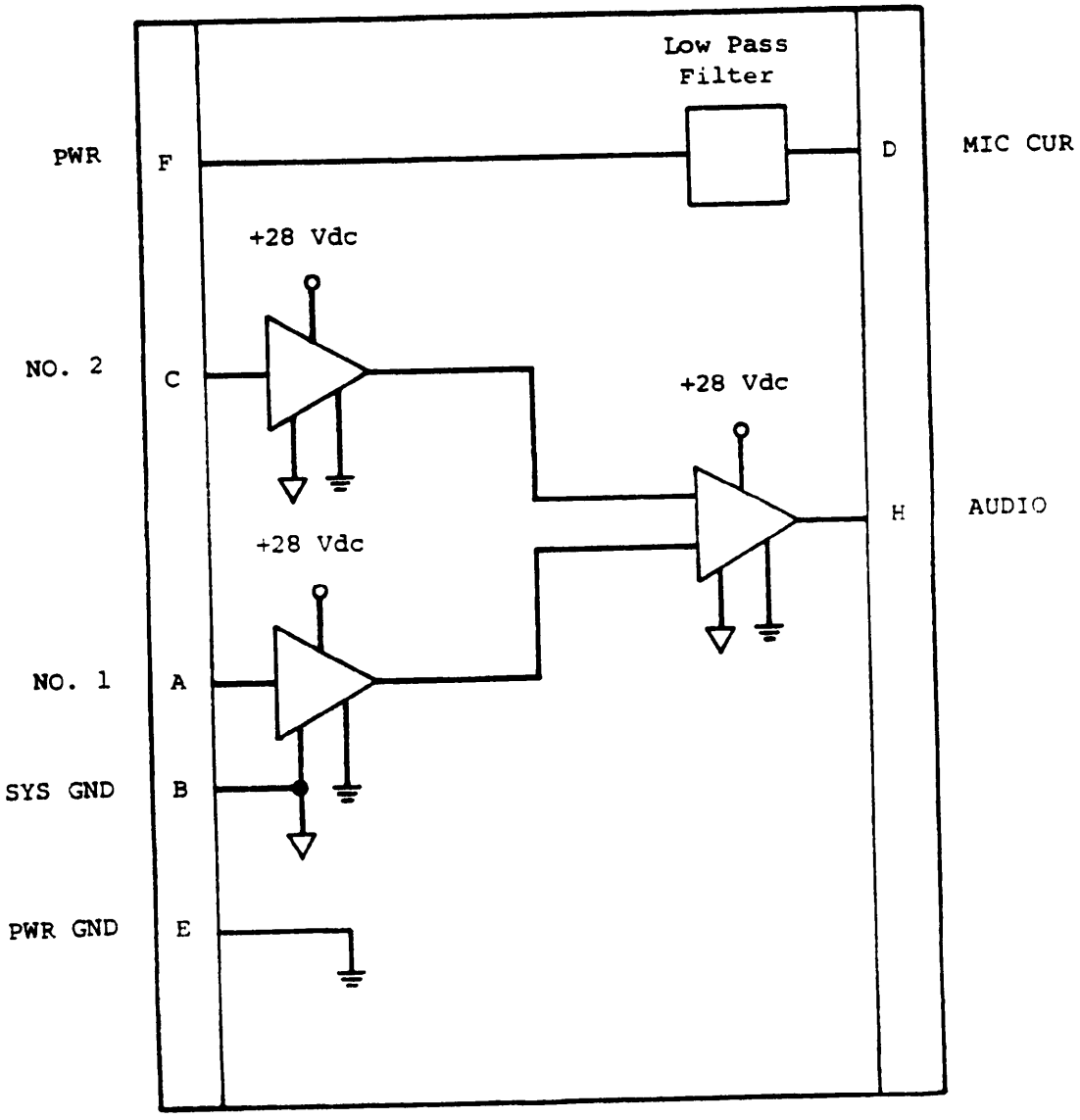
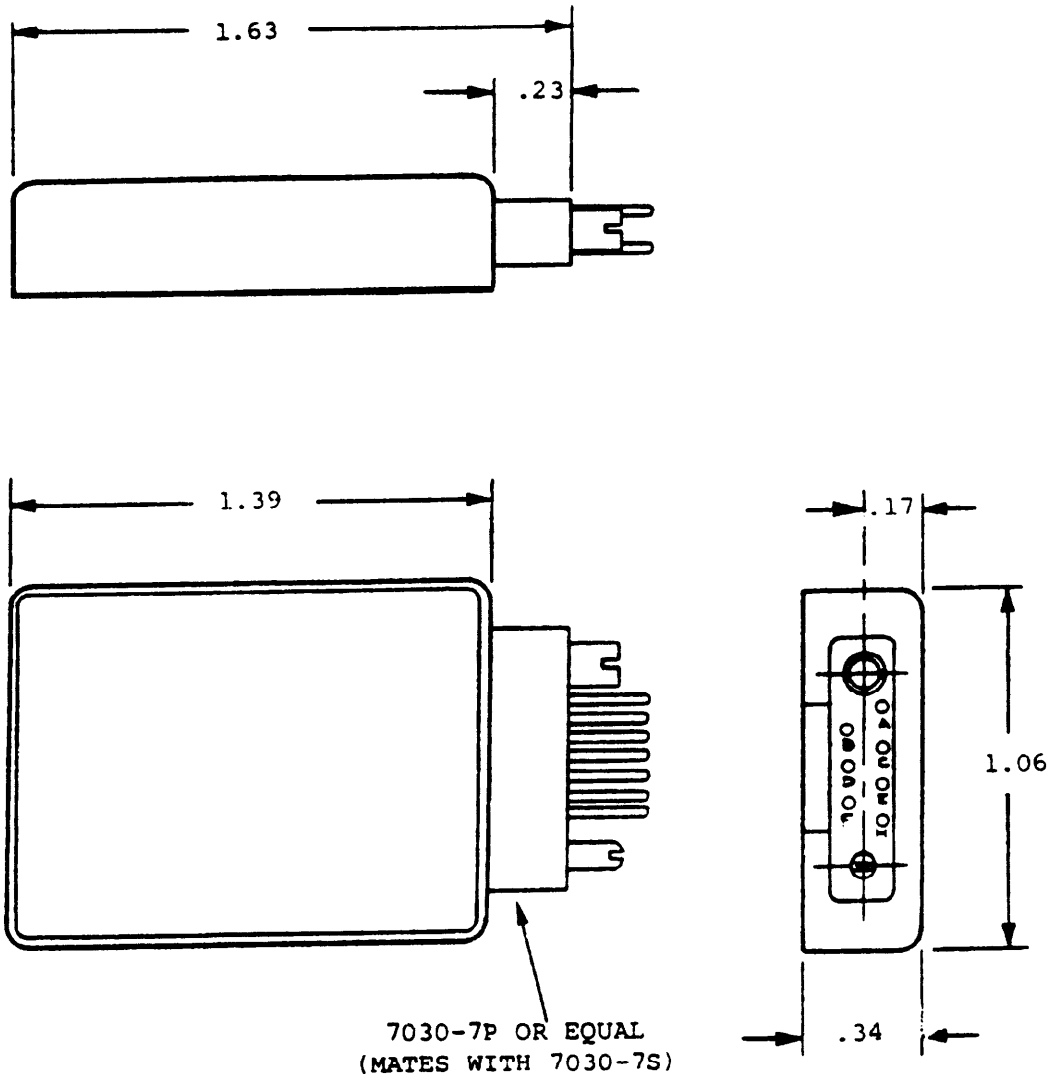


FIGURE 1. Microphone mixing module functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. Microphone mixing module outline drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/4(AS)	INTERCOMMUNICATIONS SETS, AN/AIC-14 AND AN/AIC-14A MICROPHONE MIXING MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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